

CHEONGJU INTL, Korea, Republic of

WMO: 471280

Lat: 36.717N

Lon: 127.499E

Elev: 58

StdP: 100.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-11.1	-18.9	0.7	-9.1	-16.9	0.9	-7.5	7.4	-1.9	6.6	-2.1	1.3	80	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	33.2	26.0	32.1	25.2	30.9	24.6	27.2	31.3	26.4	30.3	25.8	29.4	2.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.7	29.3	25.2	20.5	28.4	24.9	20.1	28.0	86.3	31.3	82.8	30.5	79.9	29.9	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.1	5.3	-16.5	34.9	2.9	1.4	-18.6	35.9	-20.3	36.7	-21.9	37.5	-24.0	38.6		
			-16.8	28.3	3.0	1.2	-18.9	29.2	-20.6	29.9	-22.3	30.6	-24.4	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	-2.7	0.3	5.8	12.2	18.0	22.7	25.9	26.2	21.0	14.2	6.7	-0.3
	DBStd	10.55	4.10	3.99	3.67	3.58	2.58	2.09	2.18	2.53	2.79	3.52	4.15	4.07
	HDD10.0	1266	394	273	138	18	0	0	0	0	0	8	115	320
	HDD18.3	2842	652	506	389	186	38	2	0	0	8	134	349	578
	CDD10.0	2198	0	1	7	84	248	379	492	503	331	138	16	0
	CDD18.3	733	0	0	0	2	27	131	234	245	88	6	0	0
	CDH23.3	6376	0	0	1	49	327	999	2053	2302	597	48	0	0
	CDH26.7	2129	0	0	0	4	58	290	728	913	133	3	0	0
Wind	WSAvg	2.0	1.8	2.0	2.5	2.6	2.3	1.9	2.0	1.9	1.7	1.6	1.7	1.7
Precipitation	PrecAvg	1261	20	30	44	76	92	153	288	279	153	55	43	29
	PrecMax	1830	61	95	108	192	199	369	582	826	404	199	167	58
	PrecMin	749	0	1	6	15	19	24	97	51	15	2	4	6
	PrecStd	282	16	23	26	41	40	97	120	170	107	48	35	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	14.9	21.0	27.1	30.0	32.2	35.1	35.2	31.1	26.9	21.0	13.0
		MCWB	6.4	10.2	11.9	16.6	18.7	22.2	27.0	26.6	24.0	18.9	14.8	8.6
	2%	DB	7.8	11.9	18.0	24.9	28.1	30.9	33.1	34.0	29.8	24.8	18.1	10.1
		MCWB	4.2	7.0	9.9	15.4	18.0	21.5	26.3	26.3	22.7	17.9	13.2	6.8
	5%	DB	5.9	9.8	15.9	22.8	26.9	29.8	32.0	32.9	28.1	22.9	16.1	8.1
		MCWB	2.7	5.3	9.0	14.0	17.5	21.0	25.6	25.9	21.7	16.9	11.7	5.3
	10%	DB	3.9	7.1	13.2	20.2	25.1	28.2	30.8	31.2	26.9	21.1	14.1	6.1
		MCWB	1.3	3.5	7.7	12.6	16.8	20.7	25.3	25.4	21.1	15.8	10.5	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	11.7	14.0	19.3	21.9	26.2	28.3	28.2	25.7	21.2	16.4	9.9
		MCDB	9.2	12.6	18.3	23.6	26.4	27.7	32.2	32.5	28.7	24.0	19.1	12.0
	2%	WB	4.6	7.3	11.8	17.0	20.3	24.1	27.2	27.2	24.5	19.4	14.1	7.3
		MCDB	6.6	10.6	16.1	21.7	25.2	27.6	31.1	31.7	27.6	22.7	17.2	9.4
	5%	WB	2.9	5.4	9.9	15.2	19.1	23.1	26.5	26.6	23.4	17.7	12.5	5.7
		MCDB	5.0	8.6	14.2	20.6	23.9	26.9	30.3	30.8	26.3	21.3	15.4	7.5
	10%	WB	1.5	3.9	8.3	13.8	18.0	22.2	25.9	26.0	22.3	16.5	11.0	4.1
		MCDB	3.5	6.6	12.4	19.0	23.0	25.9	29.4	29.9	25.4	20.0	13.6	5.9
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.4	12.7	13.9	12.8	10.5	7.9	8.3	10.4	12.7	11.6	10.4
		MCDBR	12.0	14.2	16.7	17.5	15.7	13.2	9.9	10.2	11.5	14.0	13.8	12.3
		MCWBR	8.7	9.7	10.1	9.2	7.2	5.5	4.3	4.2	5.6	7.8	9.1	9.3
	5% WB	MCDBR	10.4	12.7	14.2	14.1	12.1	9.5	8.4	8.8	9.2	11.6	12.0	11.3
		MCWBR	8.0	9.3	9.5	8.5	6.6	5.0	4.4	4.3	5.2	7.2	8.6	9.0
Clear-Sky Solar Irradiance	taub	0.435	0.483	0.569	0.601	0.612	0.680	0.634	0.572	0.497	0.481	0.465	0.431	
	taud	1.943	1.867	1.710	1.689	1.706	1.630	1.784	1.931	2.073	1.989	1.969	1.975	
	Ebn at Noon	705	723	699	708	709	661	689	723	757	718	670	677	
	Edh at Noon	141	172	221	238	237	256	218	184	152	150	137	129	
All-Sky Solar Radiation	RadAvg	2.49	3.24	4.23	4.91	5.54	5.07	4.12	4.28	3.99	3.55	2.50	2.22	
	RadStd	0.14	0.26	0.43	0.40	0.48	0.43	0.57	0.61	0.45	0.28	0.28	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	-0.82	N/A	N/A	N/A	N/A	N/A	+84	+46

Nomenclature: See separate page